

VisualbetaAge User Manual

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Reference

Wu, S. *, Niu, J., Yang, Y., Wang, H., Yang, J., Wu, F. 2024. First demonstration of in situ Lu–Hf dating using LA-ICP-MS/MS applied to monazite. *Journal of Analytical Atomic Spectrometry*, **39(11)**, 2703-2715

File Composition

There are four sets of files in total.

- 1. The first set “Data reduction schemes” contains DRS files for calculating Lu-Hf, Rb-Sr, Re-Os isotope ratios, consisting of 2 files (one folder plus a Python file).
- 2. The second set “User interface” contains codes for real-time display functions, consisting of two files (one is for isochronal and one is for the weighted mean).
- 3. The Lasted NIST 610 reference value file, which contain the Lu/Hf, Rb/Sr ratio data
- 4. The user Manual

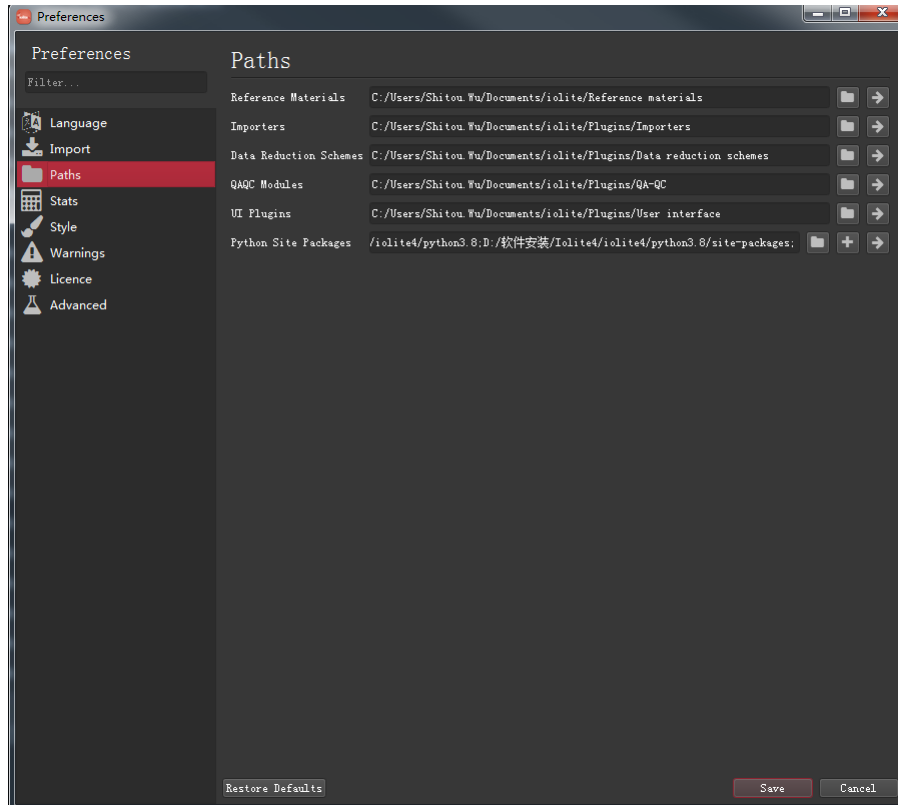
名称	修改日期	类型	大小
 Data reduction schemes	2026/8/28 19:50	文件夹	
 User interface	2026/8/28 19:44	文件夹	
 G_NIST610_2011.json	2026/6/24 10:53	JSON 文件	46 KB
 VisualbetaAge User Manual	2026/8/28 19:52	Microsoft Word ...	856 KB

名称	修改日期	类型	大小
 beta_system	2026/8/28 19:43	文件夹	
 in situ Beta system dating_V06	2026/8/28 10:03	PY 文件	17 KB

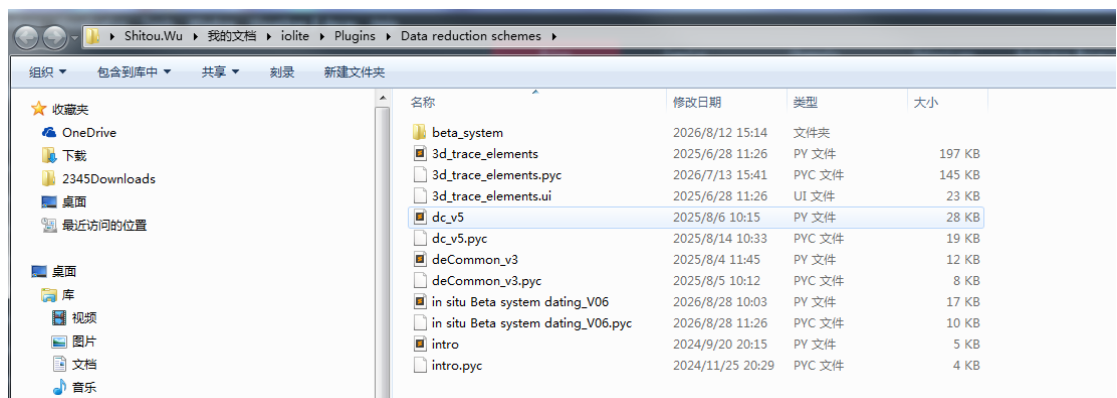
名称	修改日期	类型	大小
 livexy_20260820b	2026/8/20 15:30	PY 文件	60 KB
 weighted_mean_plot	2026/2/5 16:52	PY 文件	22 KB

Installation Procedure

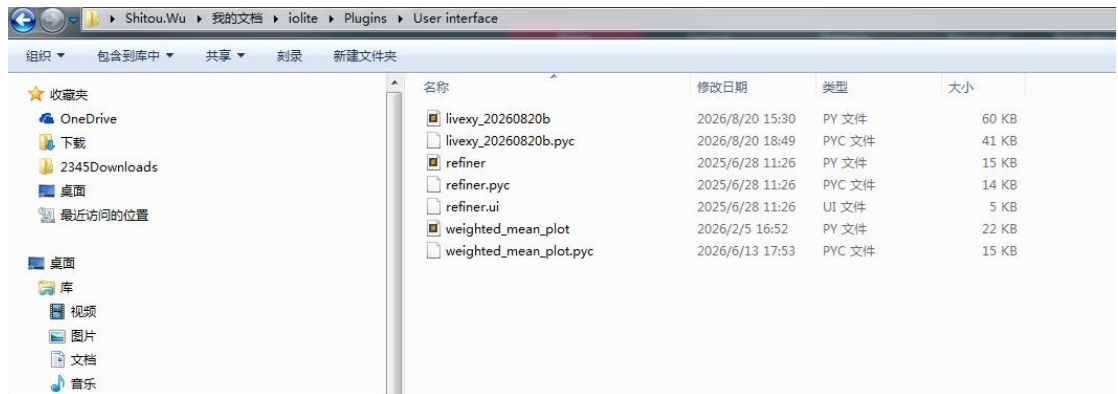
1. Open an Iolite file (either a new file or an existing data-containing file).
2. Navigate to **Session** → **Preferences** to open the Preferences dialogue box.



3. Select **Paths**, then open the folder corresponding to *Data Reduction Schemes*. Copy the files from the “Data reduction schemes” file set into this folder.



4. Still within **Paths**, open the folder corresponding to *UI Plugins*. Copy the 2 files from the second file set into this folder.



Note: After first-time installation of the code files, close Iolite and restart the software.

Operating Instructions

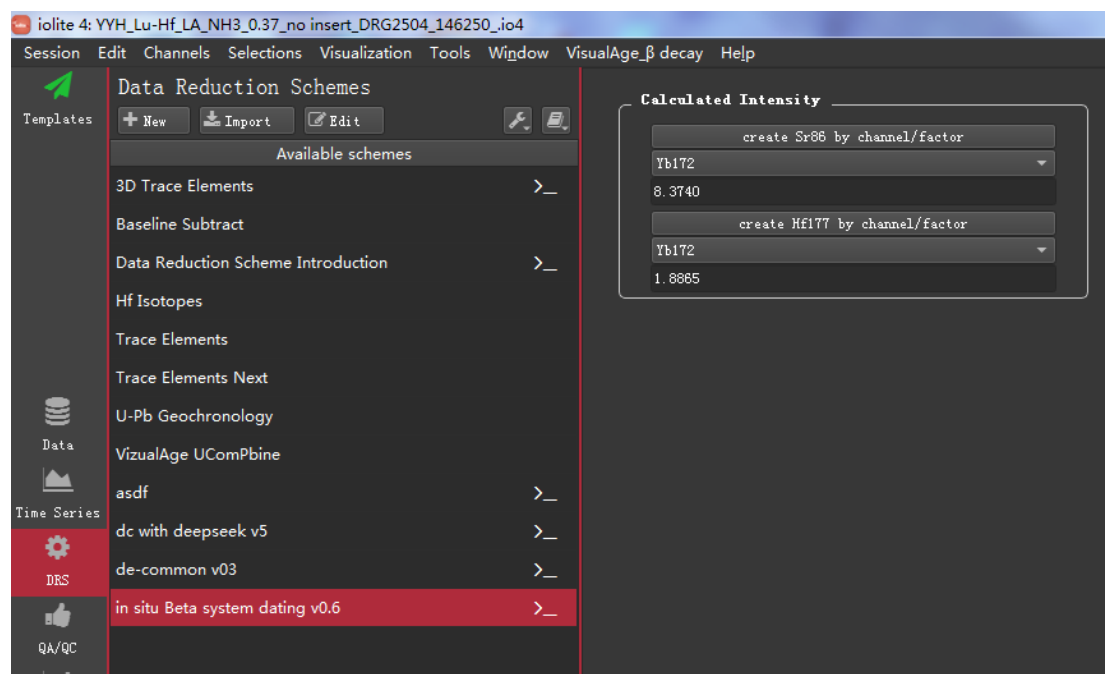
Import your dataset. Background subtraction, reference-material selection, and integration-interval selection follow the same workflows used for U-Pb data processing.

Open the DRS panel. The newly-added DRS entries will appear. Select **in-situ Beta system dating v0.6** for calculation. Meanwhile, **VisualAge_β decay** will appear in the menu bar; this function is analogous to the Visualization module for U-Pb geochronology.

Click **in-situ Beta system dating v0.6** to launch its configuration dialogue window.

Calculated Intensity Panel

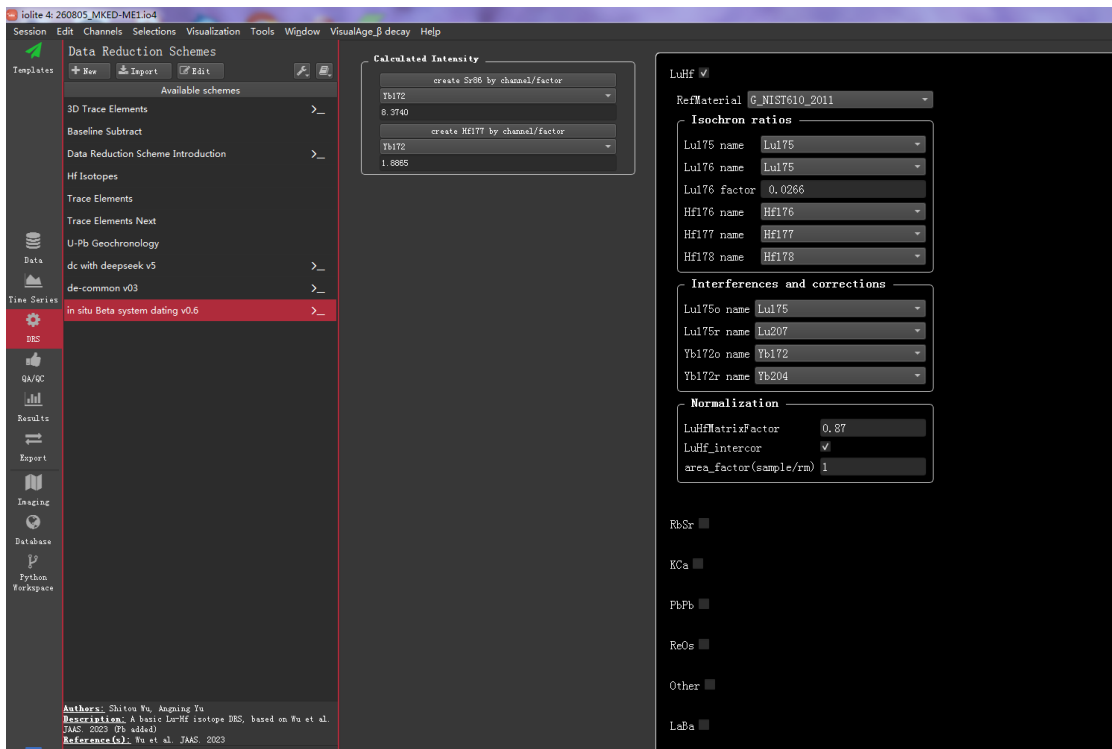
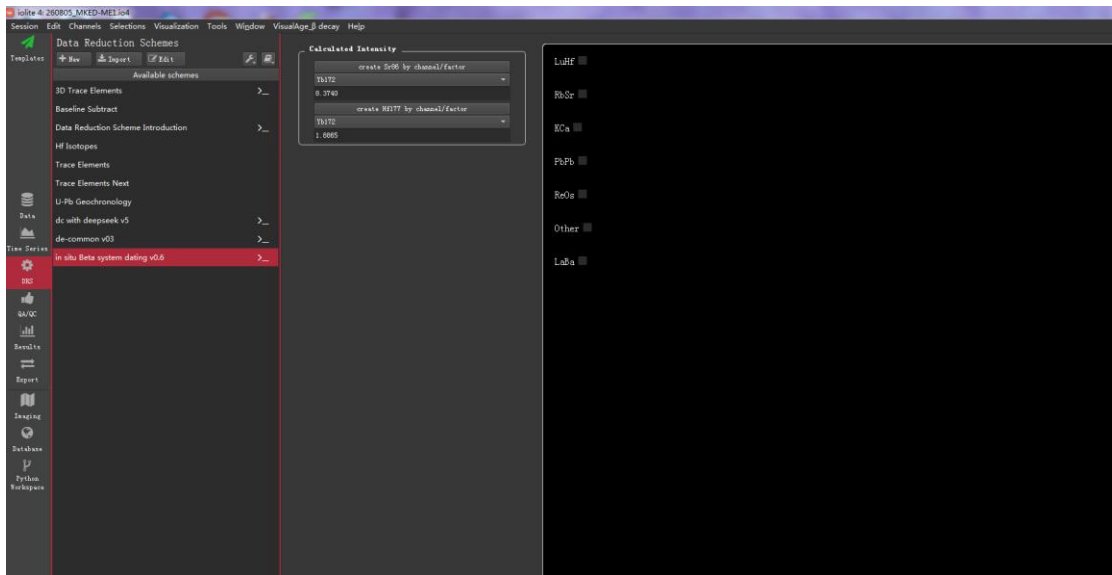
Hf177 can be calculated from Hf178, as the abundance of Hf178 is ~1.8865-times higher than that of Hf177. Click **create Hf177 by channel/factor** to perform this calculation.



Select target isotopes in the corresponding channel fields.

- The **Interferences and corrections** section is for interference correction. If no interference-monitoring signals are available, set each item to *no match*.

- The **Normalization** section displays matrix-matching correction coefficients and concentration correction coefficients (used when reference materials and unknown samples are measured using different laser spot sizes).



After completing parameter setup, click **Crunch Data** at the bottom-left corner to compute the isotope ratios. Multiple intermediate and output ratio channels will then be generated in the Channels panel.

Once ratio computation is finished via the DRS module, launch **VisualAge_β decay**. A dedicated graphical interface will pop up.

After selecting your sample group, real-time age visualization is available, operating in the same manner as for U-Pb datasets.

Define valid processing intervals, then export results using the same export workflow applied to U-Pb data. Exported data can subsequently be plotted using Isoplot or IsoplotR.

Channels					
Search...					
Yb172	Input		Yb172o_Hf177	Intermediate	
Yb204	Input		Lu175r_Lu175o	Intermediate	
Lu175	Input		Yb172r_Yb172o	Intermediate	
Lu207	Input		ILu	Intermediate	
Hf176	Input		ILu96	Intermediate	
Hf177	Input		RVb96	Intermediate	
Hf178	Input		cHf	Intermediate	
TotalBeam	Input		final_Hf176_Lu175	Output	
Yb172_CPS	Intermediate		final_Lu175_Hf176	Output	
Yb204_CPS	Intermediate		final_Lu176_Hf176	Output	
Lu175_CPS	Intermediate		final_Hf177_Hf176	Output	
Lu207_CPS	Intermediate		final_Lu176_Hf177	Output	
Hf176_CPS	Intermediate		final_Hf176_Hf177	Output	
Hf177_CPS	Intermediate		final_Lu175_Hf177	Output	
Hf178_CPS	Intermediate		final_Yb172o_Hf177	Output	
TotalBeam_CPS	Intermediate		final_Lu176_Hf176_interCor	Output	
Hf176_Lu175	Intermediate		final_Hf177_Hf176_interCor	Output	
Lu175_Hf176	Intermediate		final_Lu176_Hf177_interCor	Output	
Lu177_Hf176	Intermediate		final_Hf176_Hf177_interCor	Output	
Lu176_Hf177	Intermediate		final_Lu176_Hf177_interCor	Output	
Hf176_Hf177	Intermediate		final_Lu176_Hf176_interComCor	Output	
Lu175_Hf177	Intermediate		time	Output	
			final_Lu176_Hf176_interCor_n	Output	
			final_Hf177_Hf176_interCor	Output	
			final_Lu176_Hf177_interCor	Output	
			final_Hf176_Hf177_interCor	Output	
			final_Lu176_Hf176_interComCor	Output	
			time_n	Output	
			Lu_ppm_n	Output	
			Lu_ppm_n	Output	
			Hf_ppm_n	Output	
			Hf_ppm_n	Output	
			final_Lu176_Hf176 - final_Hf177_Hf...	Output	
			final_Lu176_Hf177 - final_Hf176_Hf...	Output	

